

JOHN ZOLTON

Patent Attorney turned ML Architect | john@zolton.xyz | github.com/JohnZolton | zolton.xyz

EDUCATION

The University of Akron School of Law - *Juris Doctor* 2023 (Honors Scholar, Dean's List)

The University of Akron - *B.S. Biomedical Engineering* 2020 (Honors, Varsity Track & Field)

EXPERIENCE

Machine Learning Architect - Skyward IT Solutions

Jan 2025 – Present

- **Led MCP initiative:** spearheaded org-wide adoption of Model Context Protocol, enabling tool-augmented LLM agents across multiple teams.
- **Scaled CMS Chat** (most popular tool) from ~50 to 7,000 daily active users by evolving it from rigid path-following to a flexible agentic system.
- Integrated MCP agents with Slack, Jira, Confluence, Linear, SQL/Vector DBs, and live Python execution.
- Designed internal tooling so subject matter experts could **author and refine agent prompts at scale**.
- Built specialized **contract research agent** with custom retrieval + task-chaining, accelerating workflows **3–5x**.
- Developed infrastructure for **LLM inference benchmarking** (Terraform, multi-cloud) with metrics like TTFT + throughput across 25+ open source models.
- Mentored new hires and interns on MCP-based agent design and production deployment.

Founder/Creator - Socratic Sensei

- Built an AI-powered study platform for law students (Next.js, tRPC, LLMs).
- Created a spaced-repetition engine that auto-generates personalized **issue-spotting essays**.
- Architected **knowledge-grounded LLM pipelines** anchored to students' notes, removing hallucinations while providing Socratic guidance.

Founder/Creator - Patense.ai

- Developed AI-driven patent analysis platform (Next.js, tRPC, Prisma, AWS Lambda).
- Achieved **2× inference performance** via vLLM, with privacy-first local + cloud deployments.
- Boosted search speed **300%** with prompt engineering + custom $O(n \log n)$ algorithmic search.

Patent Attorney - McBee, Moore & Vanik IP, LLC

2023 – 2025

- Secured patent protection across diverse technology areas.
- Managed global prosecution strategies for multiple clients.

Projects & Research

- Implemented from-scratch Rumelhart backpropagation (1986) and LeNet-5 (1998) on MNIST, demonstrating core neural network fundamentals.
- Developed EPUB/PDF → Audiobook converter: local TTS pipeline with chapter segmentation, concurrency, and custom audio handling.
- Built Gobble, an MCP-powered podcast platform: downloads, transcribes (Whisper), removes ads, and enables semantic + metadata search over transcripts with vector DBs.
- Created Hivemind, a collaborative vector DB where teams contribute solutions to accelerate onboarding and power coding agents.